Curriculum Vitae

Santiago Cifuentes

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0. Interests

→ I'm a PhD student working on the foundations of Quantum Physics, seeking to understand them through their computational aspects. Mainly, I'm interested in Quantum Computational Complexity and in grasping the difference between *randomness* as predicted by Quantum Physics and *randomness* as described through computational means (i.e. Kolmogorov Complexity). I'm looking forward to finishing my PhD in November 2026.

Currently I'm also shifting into AI Safety, especially into Agentic Foundations. Since November 2025 I've been a Research Fellow at Dovetail.

1. Current positions

- **PhD student** under the supervision of Dr. Ariel Bendersky and Dr. Santiago Figueira
  Instituto de Ciencias de la Computación, Facultad de Ciencias Exactas y Naturales (ICC, FCEyN/UBA)
  Computational Foundations of Quantum Information Theory
- **Adjunct Professor** (Profesor Adjunto)
  Facultad de Ciencias Exactas y Naturales (FCEyN, UBA)
  Algorithmic Techniques
- **Dovetail Fellow**
  Dovetail Research Group
  Working on Agentic Foundations.

2. Research experience

a) Research teams

- Member of **QUICC**[1]: *Quantum Information, Computation, and Communication*
  UBA
  2022 - Ongoing
  Working on my PhD thesis: Computational Foundations of Quantum Information Theory.
- Member of **GLyC**[2]: *Logic and Computability Group*
  UBA
  2020 - Ongoing
  Currently working on the relation between Logic and Explainable Machine Learning. Previously, I devoted myself to semi-structured knowledge representation models (specially graph-based ones) able to handle inconsistency.
- Fellow of **Dovetail**
  Dovetail Research Group
  November 2025 - Ongoing
  Research on Agentic Foundations.
- Member of the **TII** (Technology Innovation Institute) **Quantum Algorithm**'s team, at Abu Dhabi, United Arab Emirates.
  TII (Technology Innovation Institute)
  December 2023 - July 2024
  Research on quantum computational complexity, collaborating with IQIM, Caltech.

[1] https://quicc.dc.uba.ar/
[2] https://www-2.dc.uba.ar/grupinv/glyc/Home.html

b) Research projects and scholarships

- **Fellowship**
  With the Dovetail Research Group
  November 2025 - December 2026
  Research on Agentic Foundations
- **PhD Scholarship (Beca doctoral conicet)**
  With Dr. Ariel Bendersky and Dr. Santiago Figueira as directors
  April 2023 - April 2028
  Computational Foundations of Quantum Information Theory
- **PhD Scholarship (Beca doctoral conicet)**
  With Dr. Ariel Bendersky and Dr. Santiago Figueira as directors
  April 2023 - April 2028
  Computational Foundations of Quantum Information Theory
- **Master's thesis direction**
  Universidad de Buenos Aires
  (Co)Directed the following students
  - (2025) Agustin Pañale: *Emergent Misalignment when eliciting personas and fine-tuning*
  - (2025) Rafael Romani: *Understanding the theoretical limits of LLMs from the lens of computational complexity: the role of randomness inside CoT.*
  - (2024) Ezequiel Companeetz: *Improvements on the Computation of the Assymetric Shapley Values.*
  - (2024) Tomás Capdevielle: Detecting Relevant Features in Decision Diagrams.
- **Becas para la realización de la tesis de licenciatura en Ciencias de la Computación, Matemáticas y Datos**
  Director of Rafael Romani
  August 2025 - November 2025
  Understanding the theoretical limits of LLMs from the lens of computational complexity: the role of randomness inside CoT
- **Beca de Investigación en Ciencias de la Computación (BICC)**
  Mentor of Ezequiel Companeetz
  April 2024 - April 2025
  Tractable algorithms for machine learning explainability.
- **PICT-2020 Serie A**
  Query answering over graph-based databases in the presence of inconsistency
  As a student
- **Reseach assistant**
  Departamento de Computación - UBA
  2020 - 2021
  Research assistant under the supervision of Dr. María Vanina Martínez and Lic. Edwin Pin Baque.

c) Research visits

- **LISA**: *London Initiative for Safe AI*
  London
  July 2026
  Working on Agentic Foundations. More precisely, trying to develop a theoretical framework to understand convergent representations in deep neural networks.
- **LaBRI**
  Bordeaux, France
  March 2026
  Working on probabilistic and quantum automata with Laurent Bienvenu's team.
- **Technology Innovation Institute**
  Abu Dhabi, UAE
  February 2026
  Working towards understanding the quantum computational complexity of Gibbs sampling.

- **Paris Cité**
  Paris, France
  October 2025
  Working on automata and normality with Oliver Carton's team.
- **LaBRI**
  Bordeaux, France
  September 2025
  Working on probabilistic and quantum automata with Laurent Bienvenu's team.
- **Universidad Católica de Chile**
  Santiago, Chile
  December 2024
  Working on AI Explainability with Leopoldo Bertossi's team.
- **IIIA-CSIC**
  Barcelona, España
  April 2024
  Working on graph databases with María Vanina Martínez's team.
- **Universidad Católica de Chile**
  Santiago, Chile
  October 2023
  Working on AI Explainability with Miguel Romero's team.

3. Professional experience (non-research)

- **Optimization project** for local government
  September 2024 - January 2025
  Improving the routing of San Isidro's recycling trucks using combinatorial optimization techniques.
- **Full-time Engineer II** at *Medallia* (equivalent to Semi Senior)
  June 2022 - May 2023
  Worked in the *Text Analytics* team, managing service deployment, monitoring and optimization of the service pipeline
- **Full-time Engineer I** at *Medallia* (equivalent to Junior)
  February 2021 - May 2022
  Worked in the *Text Analytics* team, mainly on the optimization of the text processing pipeline

4. Teaching experience

a) University

- **Adjunct Professor** (Profesor Adjunto)
  FCEyN - UBA
  Algorithmic techniques
  May 2026 - Ongoing
- **Full-time lecturer** (Jefe de Trabajos Prácticos)
  FCEyN - UBA
  Computational Complexity
  May 2025 - August 2026
- **Full-time lecturer** (Jefe de Trabajos Prácticos)
  FCEyN - UBA
  Algorithmic techniques and graph-based algorithms (Algoritmos y Estructuras de Datos III)
  July 2022 - May 2025

- **Full-time lecturer** (Jefe de Trabajos Prácticos)
  UdeSA (Universidad de San Andrés)
  Algorithms and Data Structures
  July 2024 - December 2024
- **Assistant teacher**
  FCEyN - UBA
  Algoritmos y Estructuras de Datos III (Algorithmic techniques and graph based algorithms)
  March 2020 - June 2022

b) High school and preuniversity

- **Community education** under the project "Apoyo Escolar y Acompañamiento Educativo" of *Universidad de Buenos Aires*
  Cildañez
  Teaching math, physics and economy to highschool students, preparing them for university
  2018-2021

5. Studies and titles

a) Obtained titles

- **Licenciado en Ciencias de la Computación** (Bachelor + 2 years + Thesis, may be considered as a Master)
  Facultad de Ciencias Exactas y Naturales (FCEyN, UBA)
  2017 - 2022
  Grade: 9,52

b) Ongoing studies

- **Licenciado en Ciencias Matemáticas** (Bachelor + 2 years + Thesis, may be considered as a Master)
  Facultad de Ciencias Exactas y Naturales (FCEyN, UBA)
  95 % of the degree completed
  Current grade: 9,75

c) Competitions

- **MercadoLibre SBPO Challengue 2025**
  Combinatorial optimization competition
  Winner
- **Olimpiada Matemática Argentina** (OMA)
  Finalist
  2016 and 2017

c) Escuela de Ciencias Informáticas (Seminars of the "Computer Science School')

- **ECI 2023**
  FCEyN - UBA
  Algorithmic aspects of minor closed graph classes
- **ECI 2022**
  FCEyN - UBA
  Quantum natural computing: from simulation to algorithms
- **ECI 2021**
  FCEyN - UBA
  Quantum random number generators
- **ECI 2018**
  FCEyN - UBA
  Natural Language Processing: FastText

d) Conferences

- Alberto Mendelzon Workshop
  Santiago, Chile
  2023
- International Conference on the Principles of Knowledge Representation and Reasoning (KR)
  Virtual assistance
  2020 y 2021

6. Languages

- Spanish
  Native speaker
- English
  Advanced
  Experience talking, writing and reading
  During my time at Medallia I worked remotely with teams from different countries, talking mainly in english. I also lived 6 months in Abu Dhabi speaking in english.
- French
  Basic
  I learned it at highschool and I also lived in France for 2 months.

7. Miscellaneous

- I obtained `Dead God` on "The Binding of Isaac"

8. Publications

- Cifuentes, S. (2026). Approximate Homomorphisms and Convergent Representations in Transducers.
  ArXiv preprint arXiv:2608.20428.
- Companeetz, E., Cifuentes, S., & Abriola, S. (2026). Beyond Shapley: Efficient Computation of Asymmetric Shapley Values.
  ArXiv preprint arXiv:2606.25103.
  Student paper published at JAIIO (Jornadas Argentinas de Informática)
- Cifuentes, S. (2026). Complexity of detecting large coefficients in the Pauli basis.
  arXiv preprint arXiv:2606.19545.
  Under review in *Quantum*
- Ciancaglini, N., Cifuentes, S., Bellomo, G., Figueira, S., & Bendersky, A. (2026). Measuring the largest coefficients of a quantum state.
  ArXiv preprint arXiv:2605.00341.
  Under review in PRA (*Physical Review A*)
- Bienvenu, L., Cifuentes, S., & Gimbert, H. (2026). Characterizing normality via automata and random matrix products.
  Under review in LICS (Symposium on Logic in Computer Science).
- Companeetz, E., Cifuentes, S., & Abriola, S. (2026). Beyond Shapley: Efficient computation of asymmetric Shapley values
  In Proceedings of the Jornadas Argentinas de Informática (JAIIO).
- Cifuentes, S. (2026). General Agents Contain World Models, even under Partial Observability and Stochasticity.
  ArXiv preprint arXiv:2602.03146.
- Ciancaglini, N., Cifuentes, S., Bellomo, G., Figueira, S., & Bendersky, A. (2025). Criterios algebraicos de separabilidad y demostraciones algorítmicas de la existencia de MUBs.
  JAIIO, Jornadas Argentinas de Informática, 11(16), 130-135.

- Santesteban, M., Bruno, P., Cifuentes, S., Bellomo, G., & Bosyk, G. M. (2025). Embeddings de grafos de conocimiento basados en algoritmos cuánticos variacionales. JAIIO.
  Jornadas Argentinas de Informática, 11(16), 53-57.
- Capdevielle, T., & Cifuentes, S. (2025). Feature Relevancy, Necessity and Usefulness: Complexity and Algorithms.
  ArXiv preprint arXiv:2505.09640.
  Accepted in JAIR (Journal of Artificial Intelligence Research))
- Cifuentes, S., Wang, S., Silva, T. L., Berta, M., & Aolita, L. (2024). Quantum computational complexity of matrix functions.
  ArXiv preprint arXiv:2410.13937.
  Presented at QCTiP 2025
  Published in PRX (*Physical Review X*).
- Pardal, N., Cifuentes, S., Pin, E., Martinez, M. V., & Abriola, S. (2024). Computational Complexity of Preferred Subset Repairs on Data-Graphs
  ArXiv preprint arXiv:2402.09265
- Cifuentes, S., Bertossi, L., Pardal, N., Abriola, S., Martinez, M. V., & Romero, M. (2024) "The Distributional Uncertainty of the SHAP score in Explainable Machine Learning"
  ArXiv preprint arXiv:2401.12731
  Presented at the European Conference of Artificial Intelligence (ECAI)
- Bendersky, M. A., Bellomo, G., Ciancaglini, N., Cifuentes, S., Figueira, S. Exact algorithmic proofs of maximal mutually unbiased bases sets in arbitrary integer dimension
  ArXiv preprint arXiv:2309.12399
  Published in Quantum Information Processing Journal
- Abriola, S., Cifuentes, S., Pardal, N., & Pin, E. (2023). Data-graph repairs: the preferred approach. arXiv preprint arXiv:2304.00931.
  Presented at AMW 2023 (Alberto Mendelzon Workshop)
- Abriola, S., Cifuentes, S., Martínez, M. V., Pardal, N., & Pin, E. (2023). An epistemic approach to model uncertainty in data-graphs. International Journal of Approximate Reasoning, 160, 108948.
- Abriola, S., Martinez, M. V., Pardal, N., Cifuentes, S., & Baque, E. P. (2023). On the complexity of finding set repairs for data-graphs. Journal of Artificial Intelligence Research, 76, 721-759.
- Cifuentes, S., Soulignac, F. J., & Terlisky, P. (2023). Complexity of solving a system of difference constraints with variables restricted to a finite set. Information Processing Letters, 182, 106378.